Equations of Motion: 5-Mass Pendulum

This document contains the second-order differential equations obtained for a multiple pendulum system. Dot notation is used for time derivatives, and multi-line formatting has been applied to improve readability.

Equation 1

$$\begin{aligned}
&l_1^2 \left(m_1 + m_2 + m_3 + m_4 + m_5\right) \ddot{\theta}_1 + g l_1 \left(m_1 + m_2 + m_3 + m_4 + m_5\right) \sin\left(\theta_1\right) \\
&\quad + l_1 l_2 \left(m_2 + m_3 + m_4 + m_5\right) \sin\left(\theta_1 - \theta_2\right) \dot{\theta}_2^2 + l_1 l_2 \left(m_2 + m_3 + m_4 + m_5\right) \cos\left(\theta_1 - \theta_2\right) \ddot{\theta}_2 \\
&\quad + l_1 l_3 \left(m_3 + m_4 + m_5\right) \sin\left(\theta_1 - \theta_3\right) \dot{\theta}_3^2 + l_1 l_3 \left(m_3 + m_4 + m_5\right) \cos\left(\theta_1 - \theta_3\right) \ddot{\theta}_3 \\
&\quad + l_1 l_4 \left(m_4 + m_5\right) \sin\left(\theta_1 - \theta_4\right) \dot{\theta}_4^2 + l_1 l_4 \left(m_4 + m_5\right) \cos\left(\theta_1 - \theta_4\right) \ddot{\theta}_4 \\
&\quad + l_1 l_5 m_5 \sin\left(\theta_1 - \theta_5\right) \dot{\theta}_5^2 + l_1 l_5 m_5 \cos\left(\theta_1 - \theta_5\right) \ddot{\theta}_5 = 0
\end{aligned}$$

Equation 2

$$\begin{aligned}
&l_2^2 \left(m_2 + m_3 + m_4 + m_5\right) \ddot{\theta}_2 + g l_2 \left(m_2 + m_3 + m_4 + m_5\right) \sin\left(\theta_2\right) \\
&\quad + l_1 l_2 \left(m_2 + m_3 + m_4 + m_5\right) \cos\left(\theta_1 - \theta_2\right) \ddot{\theta}_1 + l_2 l_3 \left(m_3 + m_4 + m_5\right) \sin\left(\theta_2 - \theta_3\right) \dot{\theta}_3^2 \\
&\quad + l_2 l_3 \left(m_3 + m_4 + m_5\right) \cos\left(\theta_2 - \theta_3\right) \ddot{\theta}_3 + l_2 l_4 \left(m_4 + m_5\right) \sin\left(\theta_2 - \theta_4\right) \dot{\theta}_4^2 \\
&\quad + l_2 l_4 \left(m_4 + m_5\right) \cos\left(\theta_2 - \theta_4\right) \ddot{\theta}_4 + l_2 l_5 m_5 \sin\left(\theta_2 - \theta_5\right) \dot{\theta}_5^2 \\
&\quad + l_2 l_5 m_5 \cos\left(\theta_2 - \theta_5\right) \ddot{\theta}_5 - l_1 l_2 \left(m_2 + m_3 + m_4 + m_5\right) \sin\left(\theta_1 - \theta_2\right) \dot{\theta}_1^2 = 0
\end{aligned}$$

Equation 3

$$\begin{aligned}
&l_3^2 \left(m_3 + m_4 + m_5\right) \ddot{\theta}_3 + g l_3 \left(m_3 + m_4 + m_5\right) \sin\left(\theta_3\right) \\
&\quad + l_1 l_3 \left(m_3 + m_4 + m_5\right) \cos\left(\theta_1 - \theta_3\right) \ddot{\theta}_1 + l_2 l_3 \left(m_3 + m_4 + m_5\right) \cos\left(\theta_2 - \theta_3\right) \ddot{\theta}_2 \\
&\quad + l_3 l_4 \left(m_4 + m_5\right) \sin\left(\theta_3 - \theta_4\right) \dot{\theta}_4^2 + l_3 l_4 \left(m_4 + m_5\right) \cos\left(\theta_3 - \theta_4\right) \ddot{\theta}_4 \\
&\quad + l_3 l_5 m_5 \sin\left(\theta_3 - \theta_5\right) \dot{\theta}_5^2 + l_3 l_5 m_5 \cos\left(\theta_3 - \theta_5\right) \ddot{\theta}_5 \\
&\quad - l_1 l_3 \left(m_3 + m_4 + m_5\right) \sin\left(\theta_1 - \theta_3\right) \dot{\theta}_1^2 - l_2 l_3 \left(m_3 + m_4 + m_5\right) \sin\left(\theta_2 - \theta_3\right) \dot{\theta}_2^2 = 0
\end{aligned}$$

Equation 4

$$\begin{aligned}
&l_4^2 \left(m_4 + m_5\right) \ddot{\theta}_4 + g l_4 \left(m_4 + m_5\right) \sin\left(\theta_4\right) \\
&\quad + l_1 l_4 \left(m_4 + m_5\right) \cos\left(\theta_1 - \theta_4\right) \ddot{\theta}_1 + l_2 l_4 \left(m_4 + m_5\right) \cos\left(\theta_2 - \theta_4\right) \ddot{\theta}_2 \\
&\quad + l_3 l_4 \left(m_4 + m_5\right) \cos\left(\theta_3 - \theta_4\right) \ddot{\theta}_3 + l_4 l_5 m_5 \sin\left(\theta_4 - \theta_5\right) \dot{\theta}_5^2 \\
&\quad + l_4 l_5 m_5 \cos\left(\theta_4 - \theta_5\right) \ddot{\theta}_5 - l_1 l_4 \left(m_4 + m_5\right) \sin\left(\theta_1 - \theta_4\right) \dot{\theta}_1^2 \\
&\quad - l_2 l_4 \left(m_4 + m_5\right) \sin\left(\theta_2 - \theta_4\right) \dot{\theta}_2^2 - l_3 l_4 \left(m_4 + m_5\right) \sin\left(\theta_3 - \theta_4\right) \dot{\theta}_3^2 = 0
\end{aligned}$$

Equation 5

$$\begin{aligned}
&l_5^2 m_5 \ddot{\theta}_5 + g l_5 m_5 \sin\left(\theta_5\right) \\
&\quad + l_1 l_5 m_5 \cos\left(\theta_1 - \theta_5\right) \ddot{\theta}_1 + l_2 l_5 m_5 \cos\left(\theta_2 - \theta_5\right) \ddot{\theta}_2 \\
&\quad + l_3 l_5 m_5 \cos\left(\theta_3 - \theta_5\right) \ddot{\theta}_3 + l_4 l_5 m_5 \cos\left(\theta_4 - \theta_5\right) \ddot{\theta}_4 \\
&\quad - l_1 l_5 m_5 \sin\left(\theta_1 - \theta_5\right) \dot{\theta}_1^2 - l_2 l_5 m_5 \sin\left(\theta_2 - \theta_5\right) \dot{\theta}_2^2 \\
&\quad - l_3 l_5 m_5 \sin\left(\theta_3 - \theta_5\right) \dot{\theta}_3^2 - l_4 l_5 m_5 \sin\left(\theta_4 - \theta_5\right) \dot{\theta}_4^2 = 0
\end{aligned}$$
